

An Integrated Modelling Framework for Renewable Energy Integration in the Southern African Power Pool (SAPP)

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Abstract— Integrating renewable energy sources into regional power pools, such as the Southern African Power Pool (SAPP), presents a multifaceted challenge due to the intermittency of renewable energy resources (especially in high integration) and limited transmission infrastructure. This paper introduces an integrated modeling framework combining Open-Source Energy Modelling System (OSeMOSYS) for long-term planning, FlexiTool for managing short-term renewable variability, and Power Systems Simulation for Engineering (PSS/E) for grid validation and stability assessment. By linking strategic regional capacity planning with operational feasibility, this framework presents a comprehensive approach to optimizing the high penetration of renewable projects in the SAPP grid. The key focus of this study is on the OSeMOSYS modeling methodology, which is the primary analysis component of this framework. This also includes setting up the model and defining the parameters applicable within the SAPP context. Details of the following steps and a snapshot of some of the interconnector energy trade results are shown. Ultimately, the rationale from the modeling insights presented here can be a reference for other power pools as most regions prepare for regional integration and harmonization, coupled with market reformation, i.e., energy transitions.

Keywords—renewable integration, SAPP, OSeMOSYS, FlexiTool, transmission system optimization.

I. INTRODUCTION

The SAPP is a collaborative electricity market established in 1995. It comprises twelve Southern African Development Community (SADC) member countries [1]. The primary goal of SAPP is to improve electricity accessibility, reliability, and cost-effectiveness by enabling cross-border energy trading [2]. This regional power pool has enhanced grid resilience and efficiency, promoting shared electricity generation resources among member countries such as South Africa, Zambia, Zimbabwe, and Namibia [1].

Despite its achievements, SAPP faces considerable challenges integrating renewable energy sources such as solar,

wind, and hydro into the regional grid [3]. The region is richly endowed with renewable potential; for instance, countries like Namibia and Botswana have vast solar resources. At the same time, Mozambique and Zambia possess extensive hydropower capabilities [4]. However, the intermittent nature of these renewable sources can lead to critical grid stability and transmission issues. The existing transmission infrastructure in the SAPP was designed primarily for fossil-fuel-based generation, creating bottlenecks and limiting the extent to which renewables can be incorporated efficiently [1].

Furthermore, issues related to political coordination and regulatory alignment among SADC member countries compound these technical challenges [2]. A lack of unified governance can hinder efficient system operation and slow the implementation of transmission projects, particularly those that cross national borders (i.e., interconnectors) [5]. As such, while the SAPP framework provides a platform for integration, achieving effective renewable energy integration requires an adaptable, coordinated approach to planning and investment that can address infrastructural limitations and governance complexities [6].

The complexity of integrating high shares of renewable energy into an interconnected power pool like the SAPP requires a multifaceted approach. Traditional energy models typically focus on a single aspect of grid planning, such as economic dispatch or long-term capacity expansion [1]. While valuable, these models do not offer comprehensive insights into how strategic planning impacts short/medium-term operational reliability, nor do they effectively capture renewable energy's variability and flexibility requirements [7].

To address these limitations and as an introduction to an area of further research, this paper proposes an integrated modeling framework combining the software modeling capabilities of OSeMOSYS, FlexiTool, and PSS/E:

By integrating these tools, the proposed framework supports strategic and operational decision-making, especially for transmission utilities. Thus, it enables seamless renewables integration into the SAPP. Considering economic, technical, and policy constraints, this approach can identify optimal pathways for transmission expansion and renewable integration.

While there is a growing body of literature on renewable energy integration, existing research often fails to address both long-term planning and short-term operational challenges in a unified manner. Most studies employ macro-level optimization models focusing on economic and capacity planning or micro-level stability tools that can examine the grid in the current conditions [11]. This segmentation leaves a gap where strategic planning may overlook operational feasibility, leading to underutilized or inefficiently managed resources [11], [12].

This paper aims to close this gap by developing a comprehensive, integrated modeling framework tailored to the unique challenges of the SAPP. The primary research questions guiding this work are:

- How can an integrated approach leveraging OSeMOSYS, FlexiTool, and PSS/E address strategic planning and operational reliability for renewable integration in the SAPP?
- What insights can this framework offer regarding optimal transmission expansion and the management of renewable variability?
- How scalable is this methodology for application in other regional power pools with similar integration challenges?

This paper is structured to provide a logical flow from background and motivation to detailed methodology and expected outcomes: Section II explores existing studies and reviews various energy modeling tools and the impact of renewables on the grid; Section III discusses the proposed methodology framework with special focus on the OSeMOSYS setup, additional emphasis is also put on the FlexiTOOL and PSS/E for overall system validation; Section IV discusses the designed OSeMOSYS SAPP model with its corresponding parameters; Section V then discusses some expected results and assesses the broader implication of the methodology for SAPP.

II. LITERATURE REVIEW

A. Challenges in Integrating Renewable Energy in Power Systems

Integrating renewable energy sources into power systems poses significant challenges, particularly in regions with limited infrastructure and high energy supply variability [1]. Flexible resources such as energy storage, fast-response generation, and demand-side management are essential to maintain grid reliability. The SAPP, for example, experiences frequent curtailments due to limited storage solutions and inadequate real-time demand response capabilities [2].

Cross-border power trading, shared within the SAPP, adds further complexity to RE integration [4]. Since member countries vary widely in infrastructure maturity and resource endowment, the level of integration varies across borders, leading to transmission bottlenecks that can hinder efficient energy transfer. Additionally, governance and policy discrepancies can delay investments needed for grid improvements and hinder regional cooperation on shared projects [15].

Another critical challenge is ensuring system security and resilience amid growing VRE penetration. Increasing the balancing area can reduce aggregate variability. Diversifying supply sources across more significant regions could help stabilize the system [7]. However, this requires reliable interconnectors and coordinated real-time operations, both limited in parts of the SAPP [3]. Interconnection between countries brings benefits and risks, exposing countries to external power outages or policy changes that can ripple across borders. This means that a power trip/disturbance in Zambia at Kariba hydropower can potentially directly impact the South African grid [7].

B. Overview of Existing Modelling Approaches

Energy modeling tools are critical in understanding and optimizing power system operations. Traditional models like OSeMOSYS, Low Emissions Analysis Platform (LEAP), and Integrated MARKAL-EFOM System (TIMES) are widely used for long-term planning and policy analysis due to their ability to simulate various scenarios and policy impacts. However, these models typically lack detailed operational capabilities necessary for short-term grid management. On the other hand, software like PSS/E and DIgSILENT Power Factory excel in power flow and stability analysis but are limited in long-term economic and strategic planning. This segmentation highlights a gap in methodologies that can interlink both perspectives for comprehensive renewable integration studies.

C. Limitations of Single-Model Approaches.

Single-model approaches, whether focused on long-term planning or operational analysis, present inherent limitations when used in isolation:

- **Long-term Planning Models (e.g., OSeMOSYS):** While effective in optimizing generation and transmission investments over multi-decade horizons, these models do not consider short-term operational issues such as grid stability under fluctuating renewable inputs.
- **Operational Models (e.g., PSS/E):** These are crucial for assessing system reliability and performing power flow analysis but do not inform decisions on capacity expansion or policy impacts. Several studies have demonstrated the limitations of these approaches.

D. Advancements in Combined Modelling Frameworks

Recent advancements in energy modeling have explored integrating long-term and short-term analyses to provide a more holistic view of power systems. Studies such as in [11] have employed hybrid approaches, coupling capacity planning tools with operational models to validate proposed expansions and ensure grid reliability. This integration allows for more informed decisions, balancing economic feasibility with technical practicality.

E. Positioning of the Current Study

While the benefits of combining long-term and operational models have been recognized, few studies have specifically focused on their application in regional power pools like the SAPP [1]. The unique cross-border nature of the SAPP, coupled

with its varying levels of infrastructure maturity and renewable potential, requires a tailored approach [1]. This research positions itself as a new effort in applying an integrated modeling framework to the SAPP, aiming to bridge the gap between strategic planning and operational validation, considering the regional drive to unlock full integration of all SAPP members to increase generation supply. By leveraging the strengths of OSeMOSYS, FlexiTool, and PSS/E, this study contributes to the growing field of multi-dimensional energy modeling. It provides practical outputs for system operators, market operators, independent power producers (IPPs), governments, financiers, and donor agencies supporting several key energy infrastructure projects across the region.

III. METHODOLOGY

A. Description of OSeMOSYS for Long-term Planning

OSeMOSYS consists of equations written in the GNU MathProg language and utilizes the GNU's Linear Programming Kit (GLPK) solver to solve a system of linear equations to minimize total cost over a predefined period while meeting all energy demands. It is employed in this study to simulate multi-year capacity expansion scenarios for the SAPP, focusing on renewable energy integration and transmission network optimization. The following steps outline the setup:

- **Model Structure:** OSeMOSYS was configured to represent each SAPP member as a node, allowing for the modeling of cross-border electricity flows and interconnections.
- **Key Parameters and Assumptions:**
 - **Generation and Transmission Costs:** These are included to capture capital, operational, and maintenance costs.
 - **Renewable Energy Potential:** The data on each country's solar, wind, and hydro resources were input based on available IRENA and regional studies datasets.
 - **Policy Constraints:** Renewable energy mandates and emission targets were factored into the model to simulate realistic policy scenarios. This also included aligning this work with the current outputs from the SAPP Pool Plan.

B. OSeMOSYS inputs

OSeMOSYS comprises two main input sections: sets and parameters. Sets are the elements that make up the model. Parameters are the characteristics of those elements in the sets.

1) Sets

The sets include the following elements: emissions, technology, fuels, number of years the model will run, timeslices, modes of operation for each technology, model region, storage, and sessions in each year.

2) Scenario Design and Analysis Plan

The modeling framework supports the evaluation of multiple scenarios, each designed to test various aspects of renewable integration and transmission capacity:

- **Hydrological Shock:** Reduction of the hydropower plants capacity factors by 50% for two consecutive years in five years. This will be localized in Mozambique, Lesotho, Zambia and Zimbabwe. Angola will be included after 2035, with the commissioning of the Angola-Namibia transmission interconnector
- **Aggressive VRE:** ramping up the VRE installed capacity (MW) to 40% of the regional generation capacity by 2040.
- **Low Demand:** Considering the impact of COVID-19 and the ailing regional economic growth, this scenario assumes a 0% growth rate in the first three years and an average annual growth rate of 1.2% thereafter.
- **Deferral of Large Projects (>1,000 MW):** All generation projects above 1,000 MW are delayed by five years.
- **Flexible Generation/Storage Scenario (Optional):**
Increased deployment of storage solutions (e.g., batteries, pumped hydro) and demand-side management (DSM) to enhance system flexibility

Each scenario is analyzed through the OSeMOSYS model to determine optimal capacity expansions. Then, it is passed on to FlexiTool for flexibility assessment. The assessment will be validated in PSS/E to ensure operational feasibility. The figure below shows the simplified SAPP network illustrating the demand and generation profile per country.

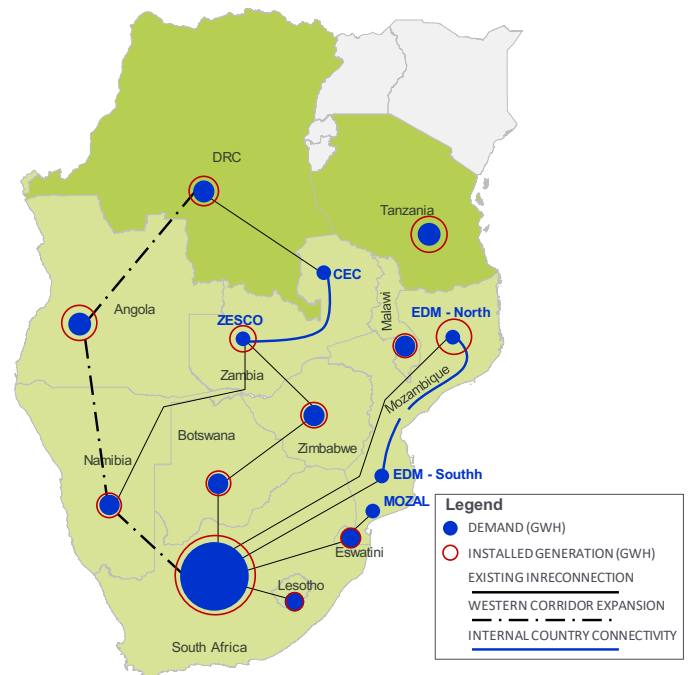


Fig. 1. Simplified modeled SAPP network

C. FlexiTool for Managing Variability

The following steps will include:

- **Data Input:** Time-series data on renewable generation profiles, demand fluctuations, and dispatchable generation capacities.
- **Flexibility Metrics:** Key outputs from FlexiTool include the grid's ramping capabilities, reserve requirements, and the impact of renewable variability on system operation domestically and regionally.
- **Simulation Outputs:** FlexiTool will provide insights into the flexibility required under various renewable penetration levels, enabling adjustments in the OSeMOSYS scenarios.

D. PSS/E for System Validation and Stability Assessment

PSS/E will analyze the outputs from the energy models by simulating power flow and stability under the proposed capacity expansion plans. The purpose is to ensure that the planned transmission and generation additions can operate reliably within the SAPP network. Key elements of this stage include:

- **Model Configuration:**
 - **SAPP Network Topology:** PSS/E replicates the existing transmission grid layout, substations, and interconnector capacities, referencing the latest project pipelines being advanced.
 - **Load Flow Analysis:** Conducted to identify potential bottlenecks and ensure that new renewable generation can be transmitted effectively without overloading the system (both domestically and regionally).
- **Validation of Results:** The findings from PSS/E are compared with outputs from OSeMOSYS and FlexiTool to confirm the feasibility of the proposed transmission expansions and renewable integration strategies.

The table below illustrates the data exchange and integration process between the models.

TABLE I. SUMMARY: INTERLINKING AND DATA TRANSFER BETWEEN MODELS

OSeMOSYS (long-term planning)	FlexiTool (short-term flexibility)	PSS/E (operational validation)
Simulates capacity expansion and interconnector investments (2024-2040)	Evaluates grid flexibility using OSeMOSYS outputs: interconnector flows and generation profiles	Tests grid stability and load flow for proposed projects using the final plans from OSeMOSYS and FlexiTool
Outputs: Generation profiles, interconnector flows, and expansion plans	Data Transfer: Time-series data from OSeMOSYS is input to assess ramping needs, reserve requirements and storage	Data Transfer: Network topology and project data are used to ensure feasibility under real-world conditions

OSeMOSYS (long-term planning)	FlexiTool (short-term flexibility)	PSS/E (operational validation)
		
<i>Feedback loop: Adjusted flexibility metrics to further refine OSeMOSYS capacity planning</i>		

IV. SAPP OSeMOSYS MODEL

This section details the SAPP OSeMOSYS model using the above format. The model is described in the subsection below.

1) SAPP Sets

The 'sets' define the physical structure of a model, usually independent from the specific scenarios that will be run. They represent the time domain [y] and time split [l], the spatial coverage [r], the technologies [t], and energy vectors to be considered, among other things.

V. SAPP MODEL OUTPUTS

The model is then solved to minimize the overall model period cost. OSeMOSYS creates large text and CSV files that provide the results described in TABLE II. These are then refined into graphs displaying different outcomes for different scenarios.

TABLE II. EXPECTED SAPP MODEL RESULTS

Name	Description
Total Emissions	Emissions will be provided per fuel type, country and rolled up for the region over the modelled time horizon.
Emission per technology per year.	Each power station's yearly emissions are calculated and shown
New capacity	Shows how much new capacity in MW was added per technology and when, e.g. when a new power station or interconnector was needed to supply demand while minimising cost.
Generation	How much electricity each technology produces per Timeslice per year in MWh.
Total cost	Total cost of the model solving.
Fuel	The amount of fuel used per Timeslice per year in MWh.
Trade	Traded energy across the region based on the energy flows (MWh) on interconnectors.

The current analysis indicates high variable energy flows between Botswana and South Africa, South Africa and Mozambique, and Botswana and Zimbabwe. These flows are influenced by the developments and increasing demands within each country.

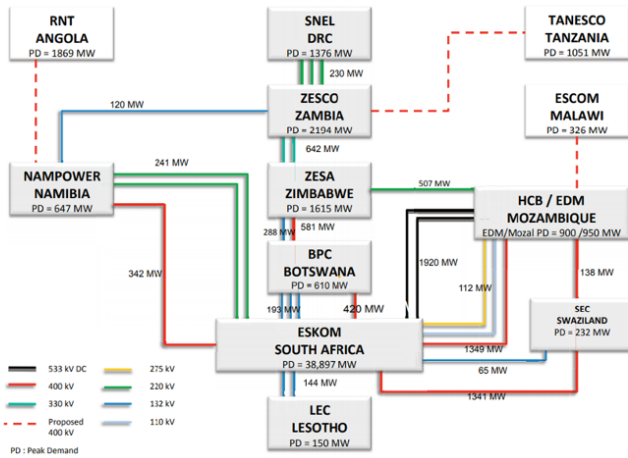


Fig. 2. Model preliminary outputs – 2030 Cross-border transfer capacities and flows

The model identifies several key generation projects for meeting the SAPP region's future energy demand while increasing renewable energy penetration. These projects were selected based on their cost-effectiveness, capacity, and ability to contribute to the regional energy mix.

Demand growth is modeled with an initial stagnation due to economic impacts followed by gradual recovery. On the supply side, renewable energy sources, such as solar and wind, are expected to increase. Cross-border interconnectors will then play a critical role in balancing electricity flows between countries, ensuring that surpluses in one region can meet deficits in another and optimizing available generation capacity.

The next steps under the study are the following:

- Completion of all years and under the different scenarios defined.
- The model will then be inputted into FlexiTool to assess the variability with a key focus on the impact of battery energy storage.
- Once the model results have been outputted, the validation process will happen in PSS/E based on some of the key projects identified per year between 2024 – 2040

While this study focuses on the SAPP, the integrated framework is designed to be adaptable for application in other regional power pools. For instance, the lessons learned from this approach can be applied to regions like the West African Power Pool (WAPP), East Africa Power Pool (EAPP), or other interconnected grids facing similar integration challenges. The ability to extend this methodology lies in its modular nature, where specific tools or data inputs can be customized to reflect the unique characteristics of other regions and countries. This scalability enhances the relevance of the study to global energy transition efforts.

It is worth noting that although the integrated modeling approach provides comprehensive coverage of strategic and operational challenges, certain limitations and data gaps may affect the robustness of the final results:

- Assumptions in Input Data: The accuracy of the findings is contingent on the quality and representativeness of input data, such as projected demand growth and generation costs. This study does not focus on demand estimation and projection but relies on existing literature.
- Model Simplifications: Integrating models may involve simplifying assumptions, such as ignoring real-time dispatch constraints in OSeMOSYS or assuming perfect foresight in FlexiTool.

These limitations can, therefore, be addressed in further studies.

VI. CONCLUSION

This paper presents an integrated modeling framework that combines long-term planning, grid stability, and operational flexibility analysis. By leveraging the combined strengths of OSeMOSYS for long-term capacity expansion, FlexiTool for short-term flexibility analysis, and PSS/E for system validation and stability assessments, the proposed methodology aims to address the gap between strategic planning and operational feasibility at a regional/power pool level. This approach also provides a comprehensive framework for optimizing renewable energy deployment while ensuring grid reliability and economic sustainability in the SAPP grid.

While this paper details more on the OSeMOSYS approach (the greater scope of the study), further empirical validation is needed to confirm its practical effectiveness. The proposal from the preliminary work done to date is that future research should focus on applying the framework to potentially other regional power pools to refine and substantiate the findings. Additionally, exploring integrating more granular, stochastic modeling elements and real-time operational data would enhance the robustness and applicability of the proposed framework. Future research should incorporate stochastic modeling with real-time operational data to improve its robustness (i.e., validating the framework using small-case studies with more advanced tools such as PLEXOS).

In conclusion, this study proposes an unexplored methodology that can serve as a template for regional power pools aiming to scale up large-scale renewable energy plants while maintaining grid stability. By addressing both long-term and short-term aspects of power system management, this framework can support energy transitions and unlock potentially new innovative business models, including climate financing.

ACKNOWLEDGMENT

The financial assistance of the National Research Foundation (NRF) in supporting this research is hereby acknowledged.

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